

SYSTEM ANALYSIS AND DESIGN

Morocco Explore

Development of a Web Platform for Discovering Tourist Destinations in Morocco

*Conception Document — Use Cases · Class Diagram · Sequence & Activity Diagrams · Database Design ·
Architecture*

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Project Type: Personal Project

Table of Contents

1. System Analysis.....	3
2. Use Case Diagram.....	6
3. Use Case Descriptions.....	7
4. Class Diagram.....	11
5. Sequence Diagrams.....	12
6. Activity Diagrams.....	16
7. Database Design.....	20
8. System Architecture.....	22
9. User Interface Structure.....	23
10. Recommended Pages.....	26
11. Final System Flow.....	27
12. Conclusion.....	28

1. System Analysis

1.1 System Actors

Morocco Explore has three main actors, each with a distinct level of access to the platform.

Visitor

A visitor is any person who accesses the website without authentication. A visitor can:

- Browse destinations.
- Browse categories.
- Search destinations.
- Filter destinations.
- View destination details.
- View images.
- View ratings and reviews.

Registered User

A registered user has all visitor privileges plus additional, personalized features. A registered user can:

- Log in.
- Manage their profile.
- Add destinations to favorites.
- Remove favorites.
- Rate destinations.
- Submit reviews.
- Manage their own reviews.

Administrator

The administrator is responsible for managing the platform's content and users. An administrator can:

- Manage destinations.
- Manage categories.
- Manage users.
- Manage reviews.
- Manage destination images.
- View platform statistics.

1.2 Functional Overview

The system can be divided into three main functional modules: Discovery (open to all visitors), the User Area (personal features for registered users), and Administration (content and user management for the administrator).

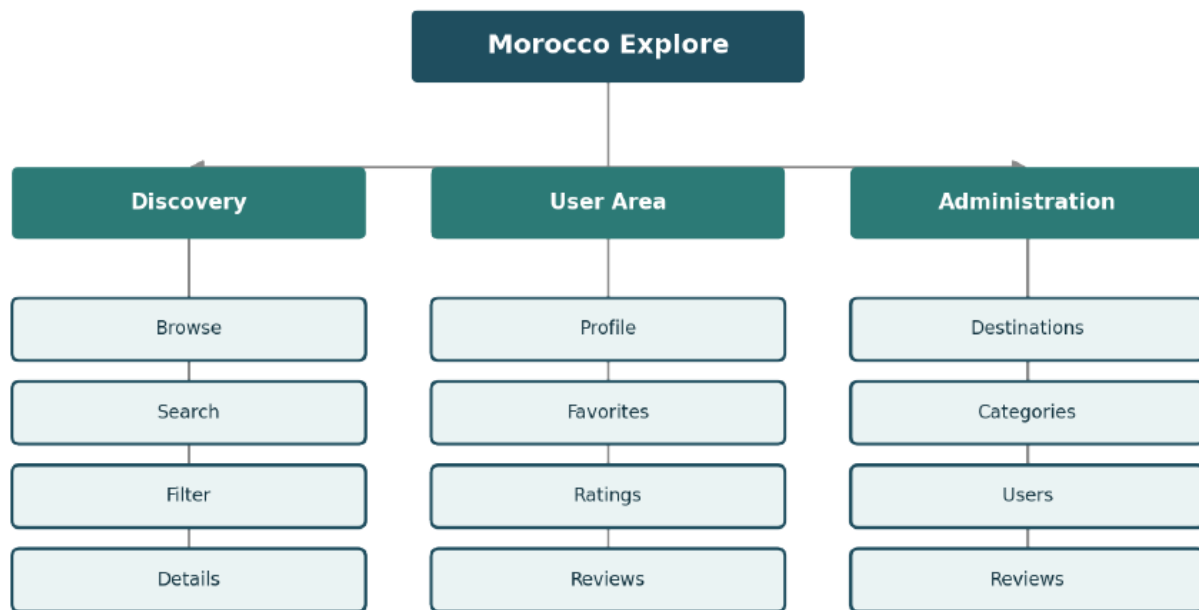


Figure 1.1 — Functional overview of Morocco Explore

1.3 Main Use Cases

The table below summarizes the main use cases per actor.

Actor	Use Case
Visitor	Browse Destinations
Visitor	Search Destinations
Visitor	Filter Destinations
Visitor	View Destination Details
Visitor	View Reviews
Visitor	Register
Registered User	Login
Registered User	Logout
Registered User	Manage Profile
Registered User	Add Favorite
Registered User	Remove Favorite
Registered User	View Favorites
Registered User	Rate Destination
Registered User	Submit Review
Registered User	Edit Review
Registered User	Delete Review
Administrator	Login
Administrator	Manage Destinations

Actor	Use Case
Administrator	Manage Categories
Administrator	Manage Users
Administrator	Manage Reviews
Administrator	Manage Images

2. Use Case Diagram

The diagram below represents the three actors (Visitor, Registered User, Administrator) and the use cases they can perform within the Morocco Explore system boundary.

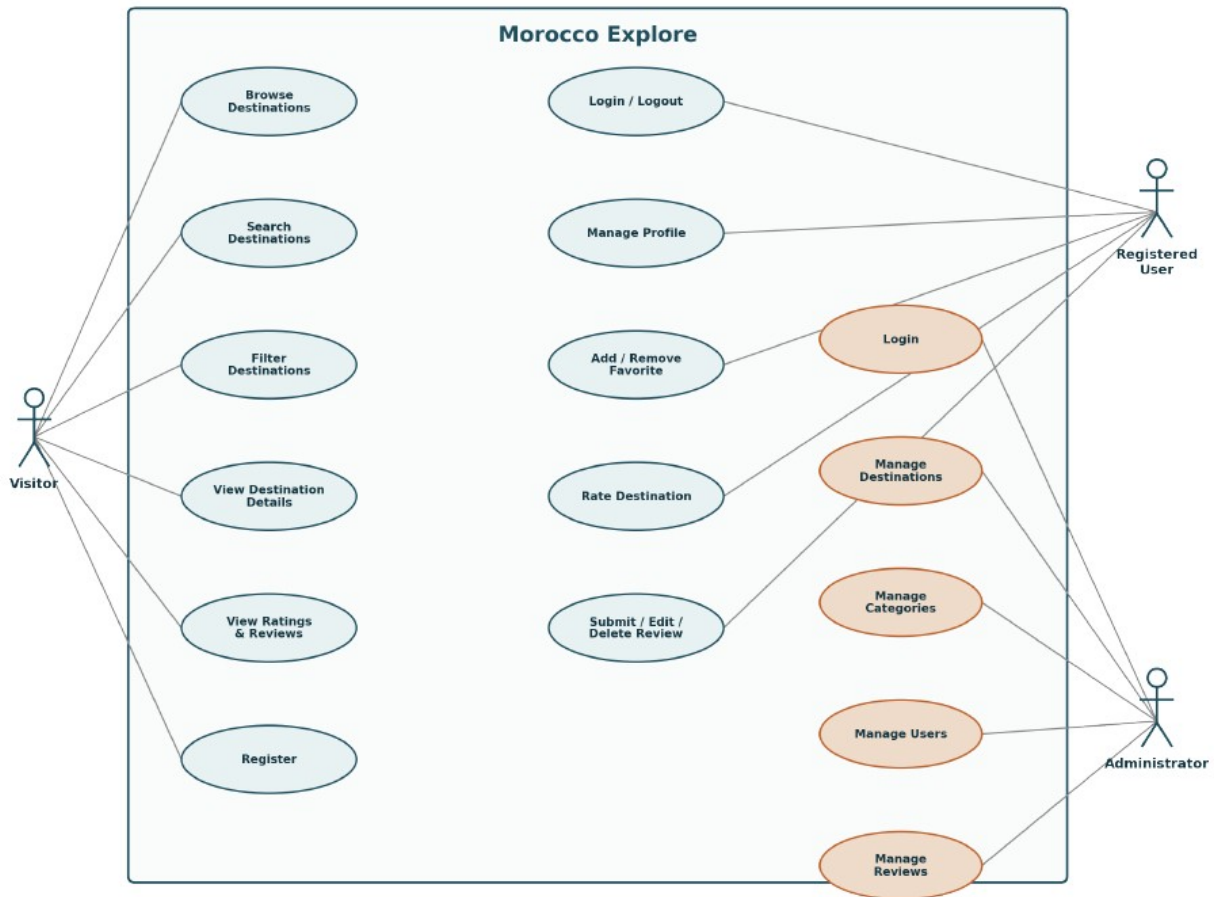


Figure 2.1 — Use case diagram of Morocco Explore

3. Use Case Descriptions

3.1 Browse Destinations

Actor: Visitor / Registered User

Precondition: The website is accessible.

Description: Allows the user to explore the available tourist destinations.

Main Flow:

1. User opens the Destinations page.
2. System retrieves available destinations.
3. System displays destination cards.
4. User can select a destination.

3.2 Search Destinations

Actor: Visitor / Registered User

Main Flow:

1. User enters a search term.
2. System receives the search request.
3. System searches destination data.
4. System displays matching destinations.

Alternative Flow: If no destination matches the query, the system displays a message indicating that no results were found.

3.3 Filter Destinations

Actor: Visitor / Registered User

The user can filter destinations by:

- Category.
- City.
- Region.
- Rating.

Main Flow:

1. User selects one or more filters.
2. System processes the selected criteria.
3. System retrieves matching destinations.
4. System displays the results.

3.4 View Destination Details

Actor: Visitor / Registered User

Main Flow:

1. User selects a destination.
2. System retrieves destination information.
3. System displays name, description, images, category, location, map, rating and reviews.

3.5 Register

Actor: Visitor

Main Flow:

1. Visitor opens the registration page.
2. Visitor enters required information.
3. System validates the information.
4. System checks email uniqueness.
5. System creates the account.
6. Registration succeeds.

3.6 Login

Actor: Registered User / Administrator

Main Flow:

1. Actor enters email and password.
2. System validates credentials.
3. System authenticates the actor.
4. System redirects the actor to the appropriate dashboard/page.

3.7 Manage Profile

Actor: Registered User

The user can:

- View profile.
- Edit profile.
- Change password.

3.8 Manage Favorites

Actor: Registered User

Add Favorite

1. User opens a destination.
2. User clicks Add to Favorites.
3. System checks whether it is already saved.
4. System creates the favorite.

5. System confirms the operation.

Remove Favorite

1. User opens favorites.
2. User selects a destination.
3. User clicks remove.
4. System deletes the favorite.

3.9 Rate Destination

Actor: Registered User

Main Flow:

1. User opens a destination.
2. User selects a rating.
3. System validates the rating.
4. System saves the rating.
5. The destination's average rating is updated.

3.10 Submit Review

Actor: Registered User

Main Flow:

1. User opens a destination.
2. User enters a review.
3. User submits the review.
4. System validates the content.
5. System stores the review.
6. The review is displayed according to the platform's moderation rules.

3.11 Manage Destinations

Actor: Administrator

The administrator can:

- Add destination.
- View destinations.
- Edit destination.
- Delete destination.
- Publish/unpublish destination.

Destination information includes:

- Name, description, city, region, category.
- Location (latitude / longitude).
- Images.

- Additional information.

3.12 Manage Categories

Actor: Administrator

The administrator can:

- Create category.
- Edit category.
- Delete category.
- View categories.

3.13 Manage Users

Actor: Administrator

The administrator can:

- View users.
- Search users.
- View user information.
- Activate/deactivate accounts.
- Delete accounts.

3.14 Manage Reviews

Actor: Administrator

The administrator can:

- View reviews.
- Approve reviews.
- Hide reviews.
- Delete inappropriate reviews.

4. Class Diagram

4.1 Main Classes

The main classes of the system are: **User**, **Category**, **Destination**, **Image**, **Favorite**, **Review** and **Rating**.

Rather than creating a separate *Administrator* table, the Administrator is modeled as a **User with an administrative role** (`role = USER | ADMIN`). This reduces duplication and keeps authentication logic simple, while still allowing role-based access control throughout the application.

4.2 Class Structure and Relationships

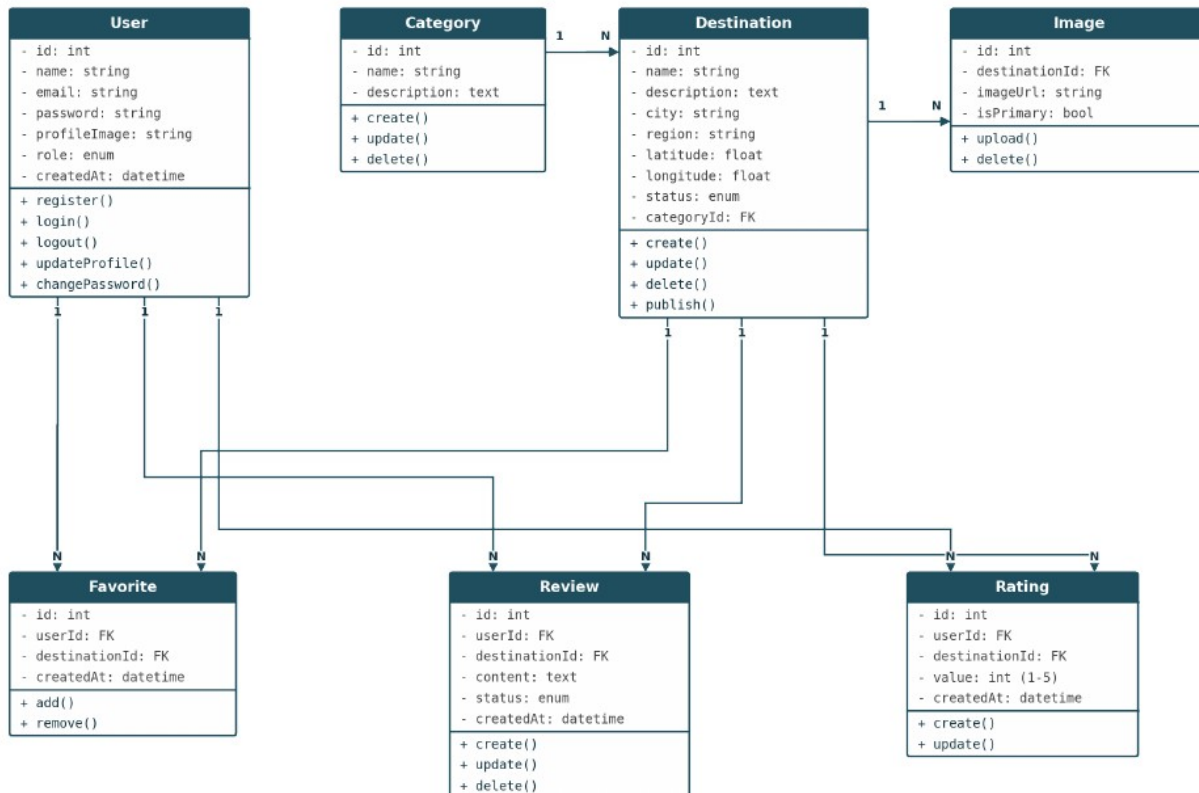


Figure 4.1 — Class diagram of Morocco Explore

Relationship summary

- **Category (1) — Destination (N)**: one category groups many destinations.
- **Destination (1) — Image (N)**: one destination can have multiple images.
- **User (1) — Favorite (N) and Destination (1) — Favorite (N)**: Favorite is an association entity between User and Destination.
- **User (1) — Review (N) and Destination (1) — Review (N)**: a user can write several reviews; a destination can receive several reviews.
- **User (1) — Rating (N) and Destination (1) — Rating (N)**: a user can rate several destinations; a destination can receive several ratings.

5. Sequence Diagrams

The sequence diagrams below illustrate the interactions between the actor, the interface, the server (application layer) and the database for the system's key operations.

5.1 User Registration

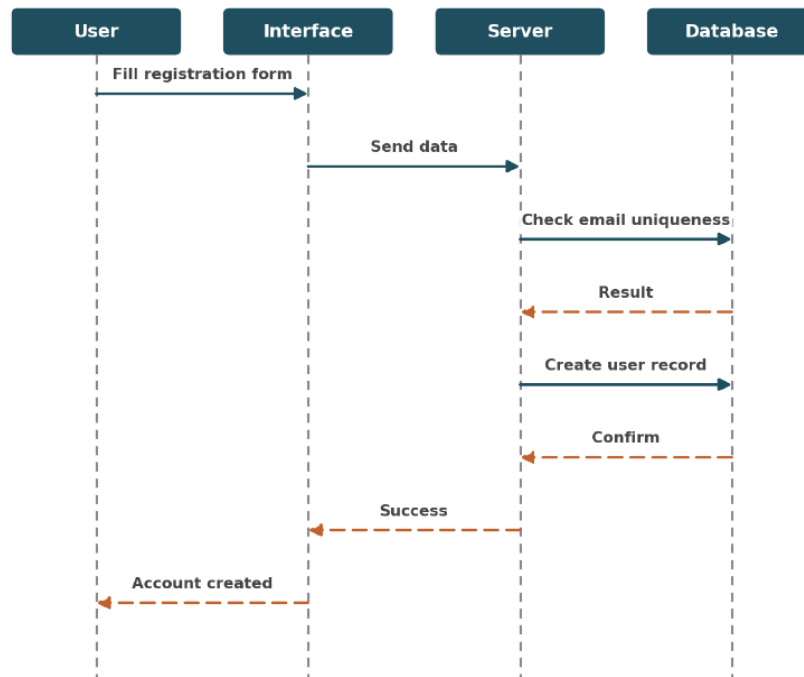


Figure 5.1 — Sequence diagram: User Registration

5.2 User Login

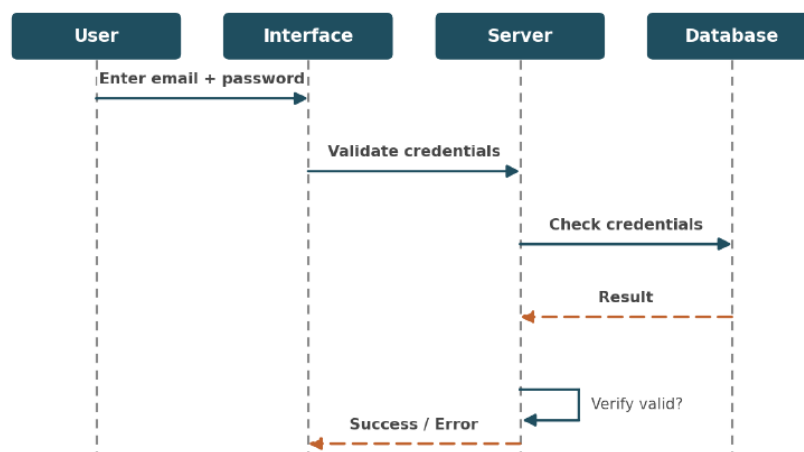


Figure 5.2 — Sequence diagram: User Login

5.3 Search Destination

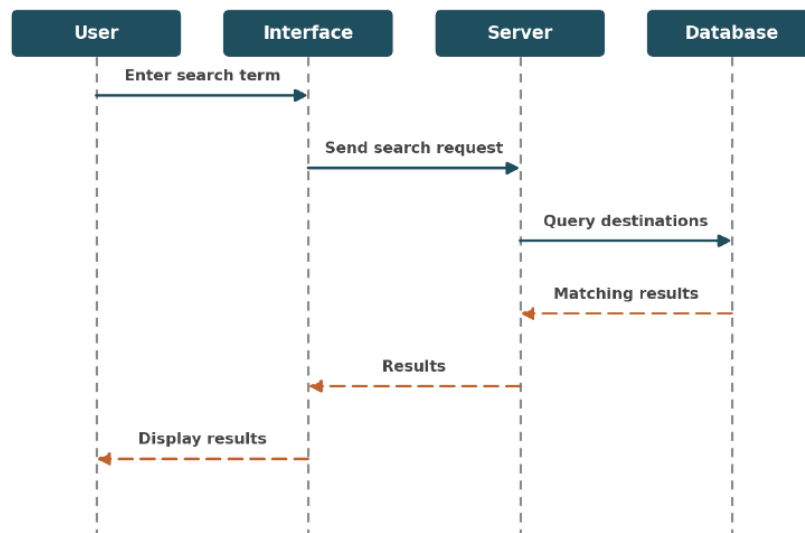


Figure 5.3 — Sequence diagram: Search Destination

5.4 View Destination Details

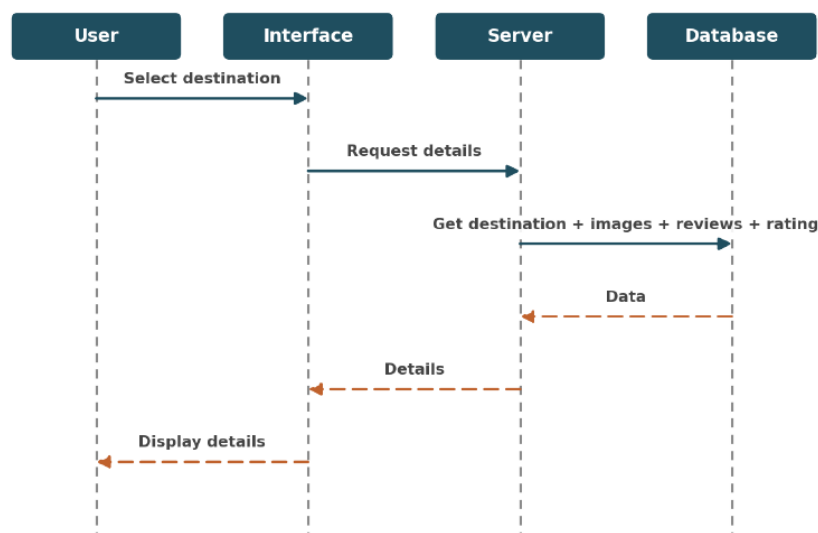


Figure 5.4 — Sequence diagram: View Destination Details

5.5 Add Favorite

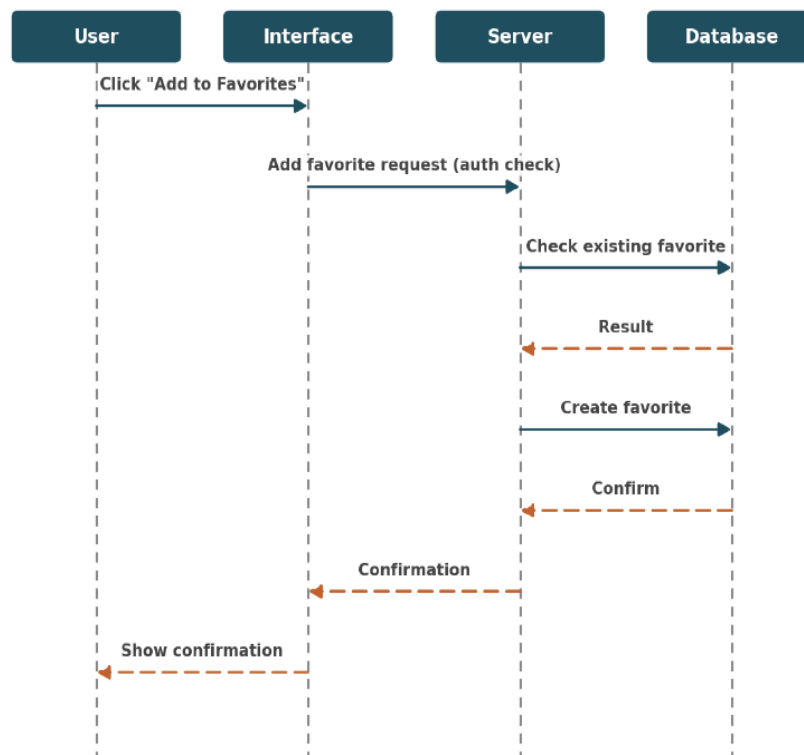


Figure 5.5 — Sequence diagram: Add Favorite

5.6 Submit Review

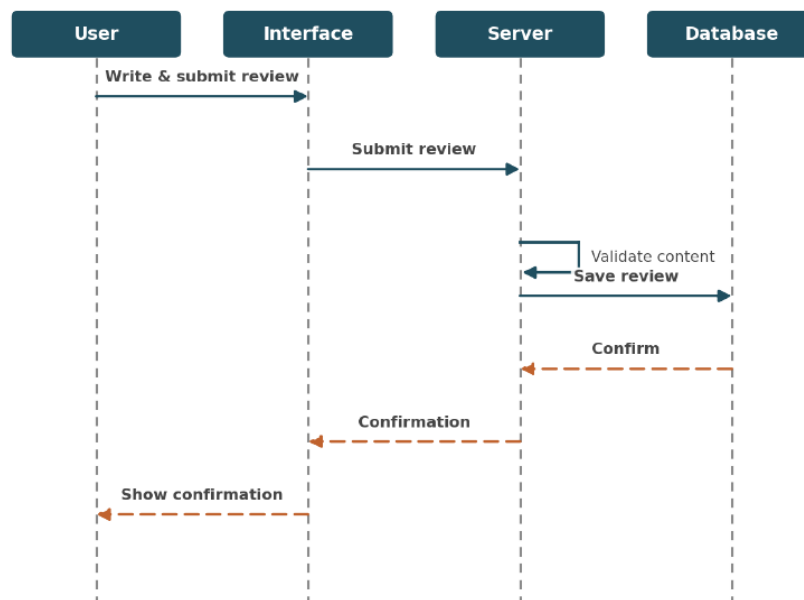


Figure 5.6 — Sequence diagram: Submit Review

5.7 Administrator Adds a Destination

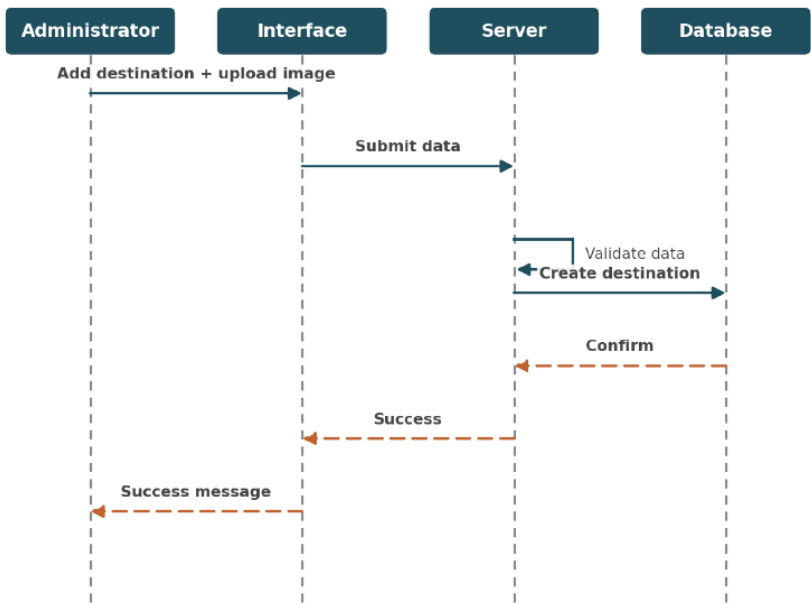


Figure 5.7 — Sequence diagram: Administrator Adds a Destination

6. Activity Diagrams

The activity diagrams describe the step-by-step logic and decision points behind the system's main workflows.

6.1 Browse / Search Destination

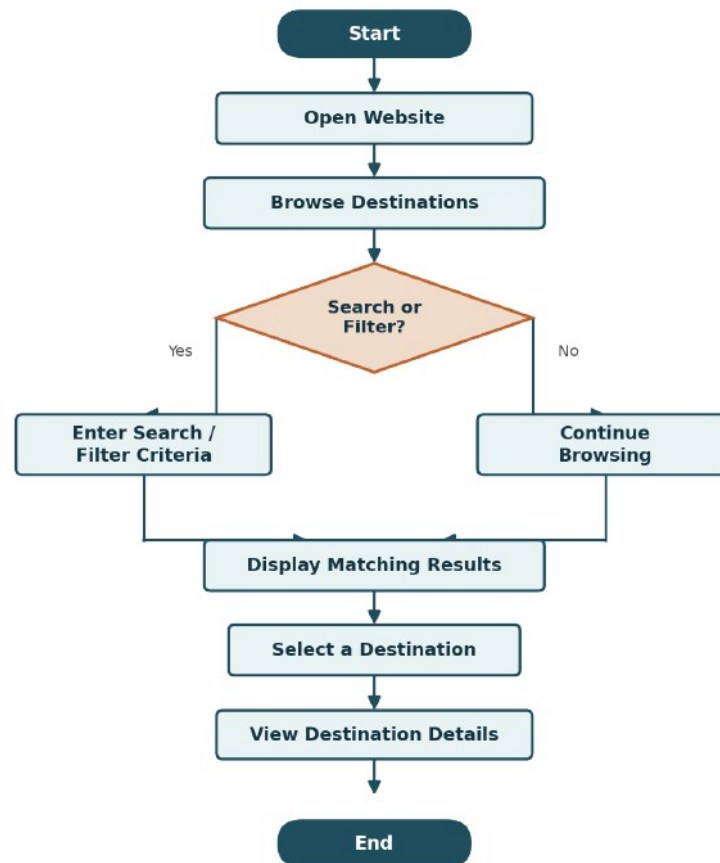


Figure 6.1 — Activity diagram: Browse / Search Destination

6.2 User Login

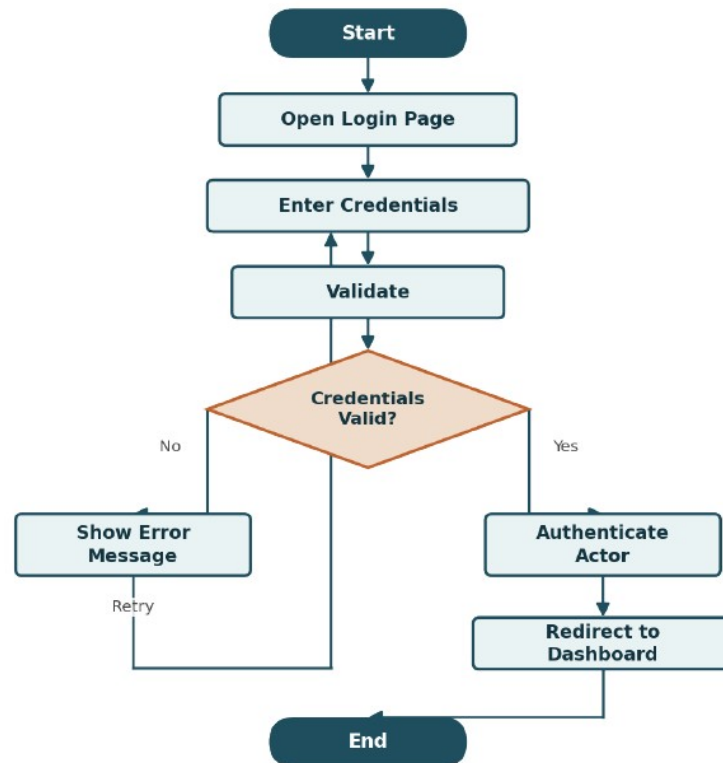


Figure 6.2 — Activity diagram: User Login

6.3 Add Favorite

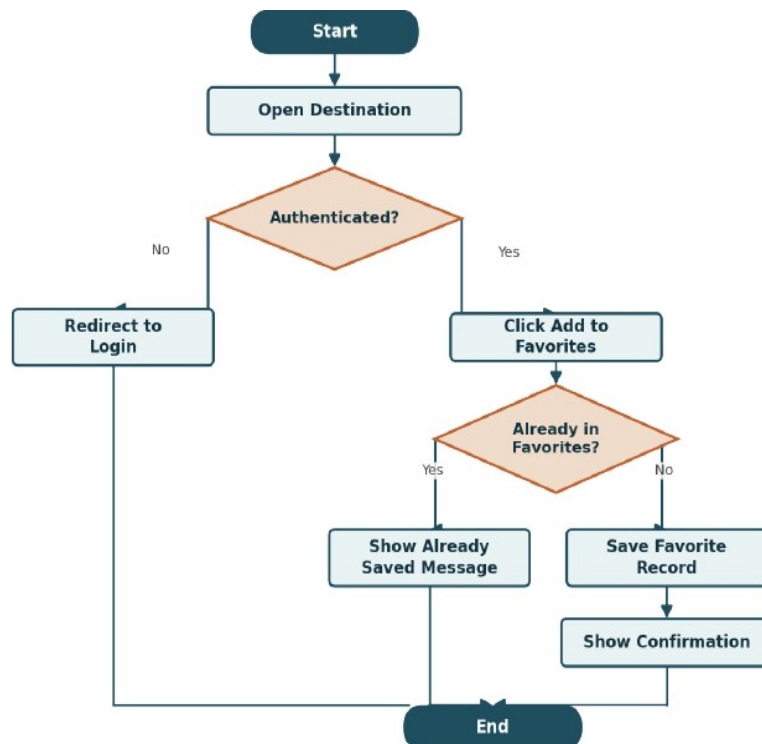


Figure 6.3 — Activity diagram: Add Favorite

6.4 Submit Review

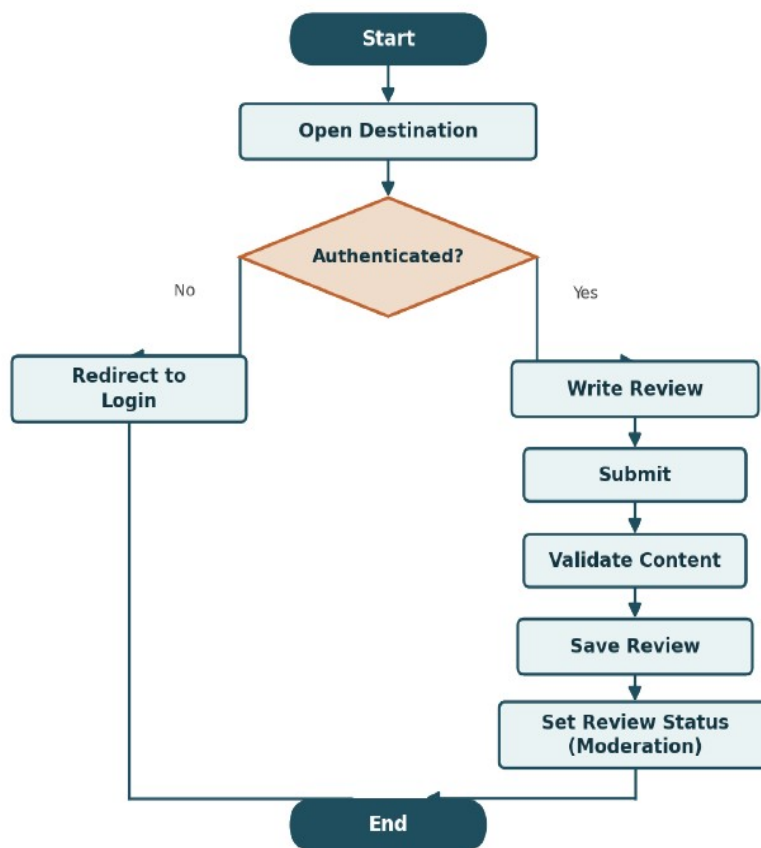


Figure 6.4 — Activity diagram: Submit Review

6.5 Administrator Adds a Destination

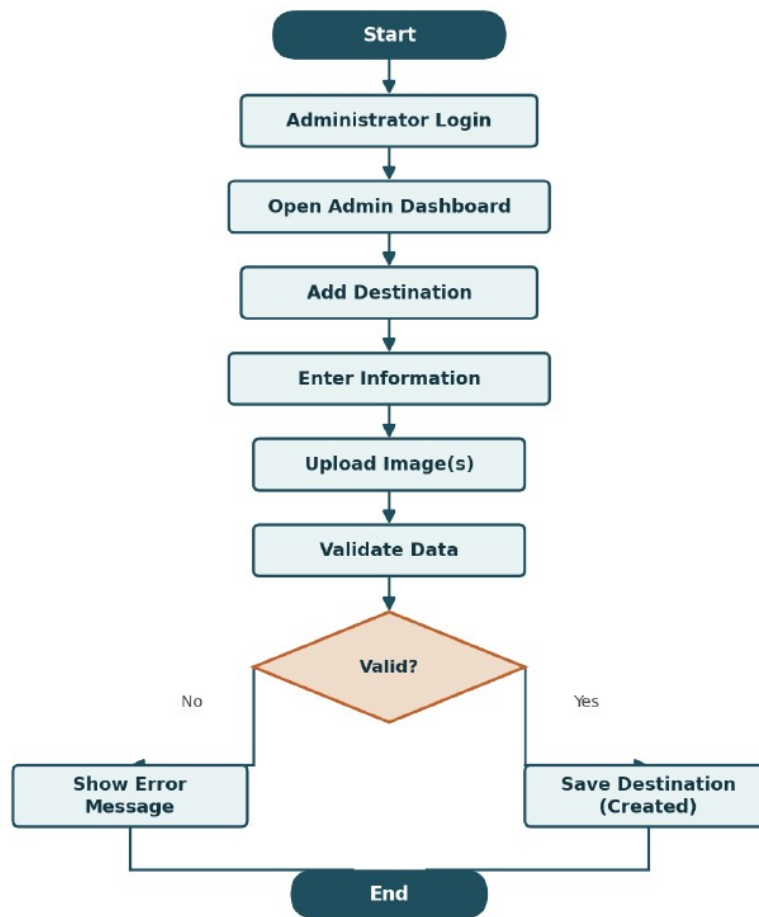


Figure 6.5 — Activity diagram: Administrator Adds a Destination

Figure 6.5 — Activity diagram: Administrator Adds a Destination

7. Database Design

7.1 Entities

The database will contain the following main entities:

- Users
- Categories
- Destinations
- Images
- Favorites
- Reviews
- Ratings

7.2 – 7.3 Entity Relationship Diagram (ERD)

The Entity-Relationship Diagram below shows every entity, its primary/foreign keys, and the cardinalities that connect them.

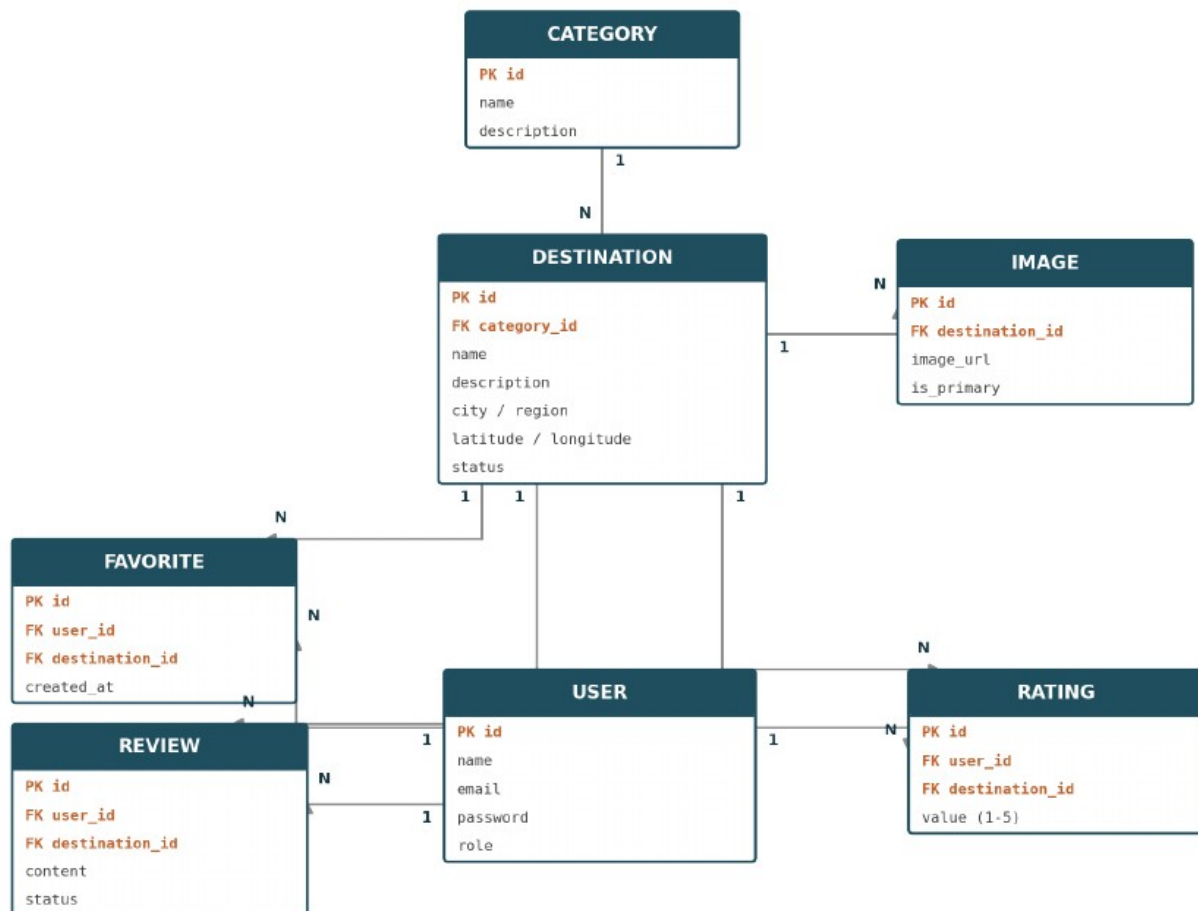


Figure 7.1 — Entity-Relationship Diagram (ERD)

- **Category (1) → Destination (N)**: one category can contain many destinations.

- **Destination (1) → Image (N):** one destination can have multiple images.
- **User (1) → Favorite (N) and Destination (1) → Favorite (N):** Favorite is the association entity between User and Destination.

7.4 Logical Data Model (MLD)

The proposed logical data model (relational schema) is as follows:

USERS (id PK, name, email UNIQUE, password, profile_image, role, created_at, updated_at)
CATEGORIES (id PK, name, description, created_at, updated_at)
DESTINATIONS (id PK, category_id FK, name, description, city, region, latitude, longitude, status, created_at, updated_at)
IMAGES (id PK, destination_id FK, image_url, is_primary, created_at)
FAVORITES (id PK, user_id FK, destination_id FK, created_at)
REVIEWS (id PK, user_id FK, destination_id FK, content, status, created_at, updated_at)
RATINGS (id PK, user_id FK, destination_id FK, value, created_at, updated_at)

7.5 Database Constraints

- **Users.email** → **UNIQUE**, so that no two users share the same email address.
- **Favorites (user_id, destination_id)** → **UNIQUE**, so a user cannot save the same destination to favorites twice.
- **Ratings (user_id, destination_id)** → **UNIQUE**, so a user has at most one rating per destination.
- **Ratings.value** → must respect $1 \leq \text{value} \leq 5$.

8. System Architecture

Morocco Explore follows a classic **three-layer (three-tier) architecture**, separating presentation, application logic and data storage.

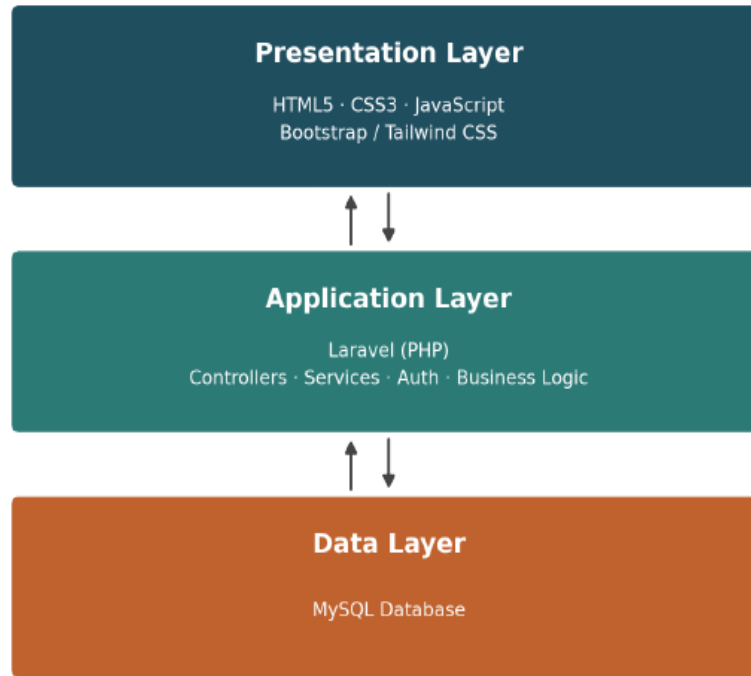


Figure 8.1 — Three-layer system architecture

8.1 – 8.2 Layer Responsibilities

Presentation Layer (HTML / CSS / JS, Bootstrap or Tailwind)

- Pages and forms.
- Navigation and destination cards.
- Responsive interface.
- User interactions.

Application Layer (Laravel / PHP)

- Authentication.
- Business logic and validation.
- Search and filtering.
- Favorites, reviews and ratings processing.
- Administration workflows.

Data Layer (MySQL)

- Storing users, destinations, categories.
- Storing images, reviews, ratings and favorites.

9. User Interface Structure

The platform is organized into three navigational areas, matching the three system actors.

Public Interface

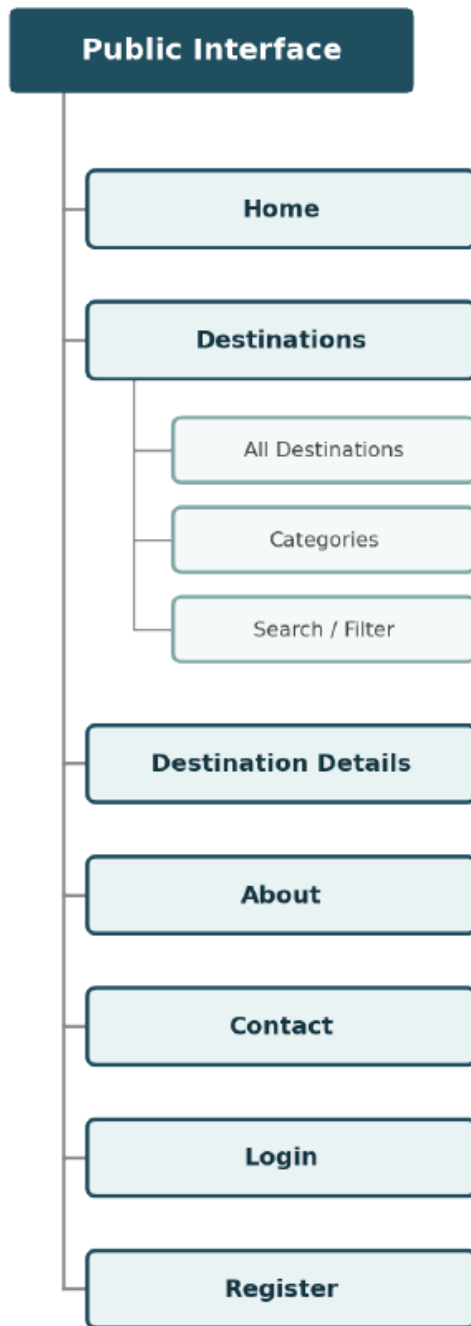


Figure 9.1 — Public interface navigation structure

Registered User Interface

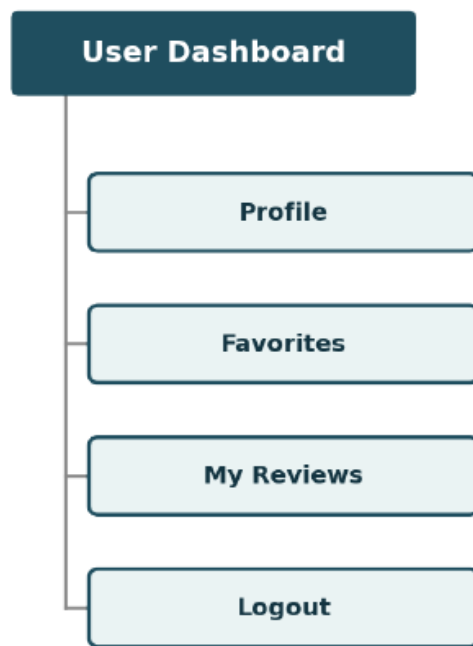


Figure 9.2 — User dashboard navigation structure

Administrator Interface

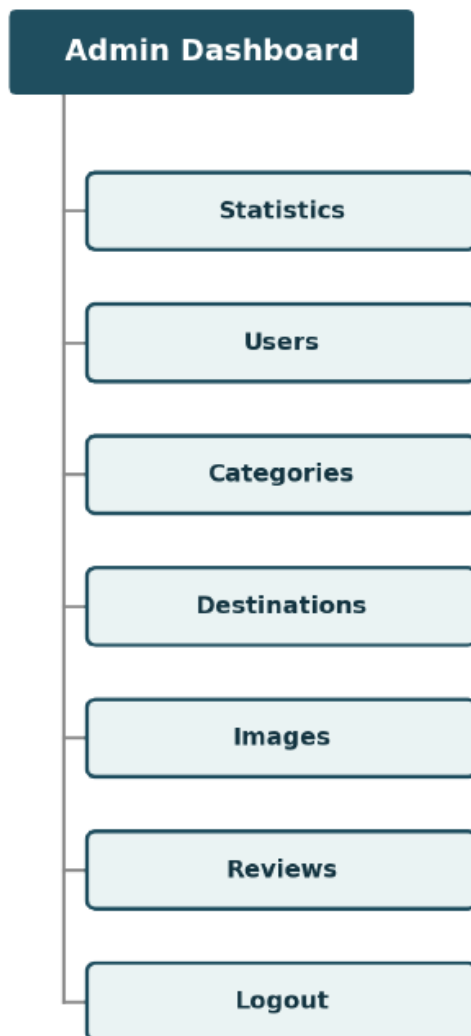


Figure 9.3 — Admin dashboard navigation structure

10. Recommended Pages

To keep the project clear and complete, the following pages are recommended for the first version of the platform:

Public	Registered User	Administrator
Home	Dashboard	Admin Dashboard
Destinations	Profile	Manage Destinations
Destination Details	Favorites	Add / Edit Destination
Categories	My Reviews	Manage Categories
Search Results		Manage Users
About		Manage Reviews
Contact		
Login		
Register		

11. Final System Flow

Registered User Journey

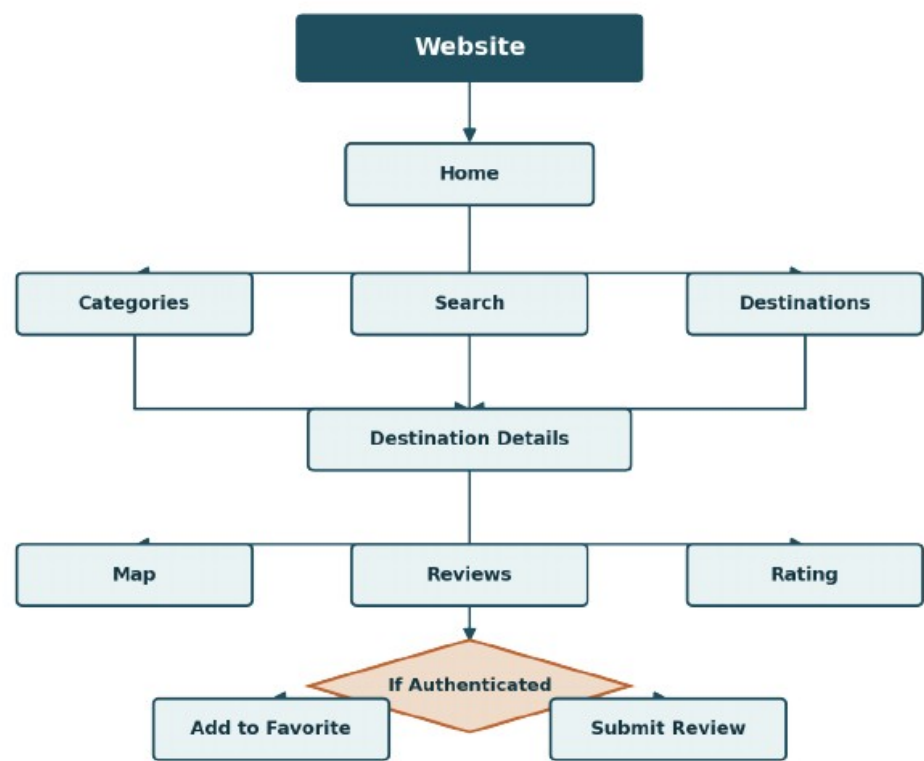


Figure 11.1 — End-to-end flow for a regular user

Administrator Journey

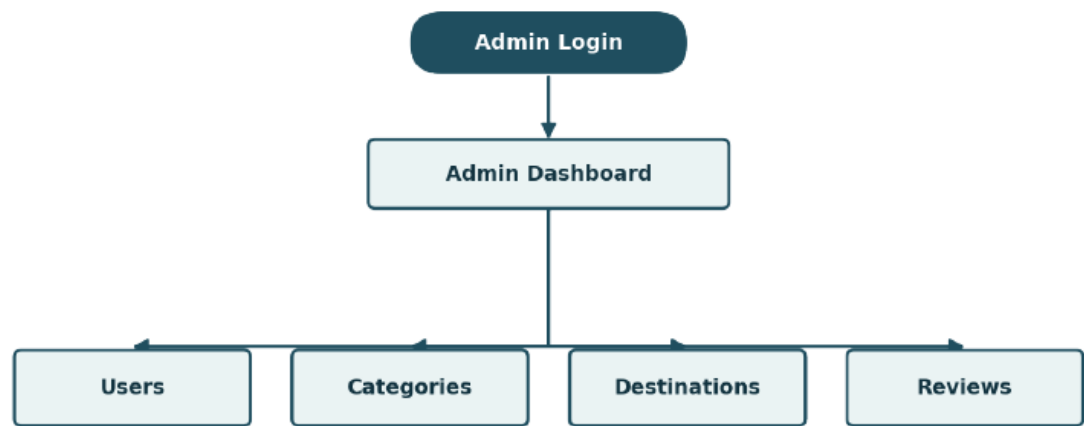


Figure 11.2 — End-to-end flow for the administrator

12. Conclusion

This document establishes the complete System Analysis & Design (conception) of Morocco Explore, built directly on top of the project's Software Requirements Specification (SRS):

- Actors and their use cases.
- Detailed use case descriptions.
- Class diagram and class relationships.
- Sequence diagrams for the main scenarios.
- Activity diagrams for the main workflows.
- Database entities, Entity-Relationship Diagram (ERD) and logical data model (MLD).
- Database constraints.
- Three-layer system architecture.
- User interface structure and recommended pages.

With actors, use cases, the class diagram, the ERD and the sequence/activity diagrams now fully specified and drawn, the project is ready to move into the implementation phase.

Recommended technology stack: **Laravel + MySQL** for the back-end and database, and **HTML / CSS / JavaScript with Bootstrap or Tailwind CSS** for the front-end. The final choice of technologies can still be adjusted before development begins.